

Novel Prep Math Placement Test

Algebra I – Calculus Readiness

Name: _____

Date: _____

Directions: You can only use your calculators in Part 5. Choose the best answer for each question. This placement test is designed to help determine readiness for Algebra I, Geometry, Algebra II, Precalculus, or Calculus-level coursework.

Part I — Foundations and Algebra Skills

- 1 A factor of $x^2 - 8x + 15$ is
- A) $x - 3$
 - B) $x + 3$
 - C) $x + 5$
 - D) $x - 8$
 - E) $x - 15$
- 2 One solution to $2x^2 - 5x - 3 = 0$ is
- A) -3
 - B) -1
 - C) 0
 - D) 1
 - E) 3
- 3 $\frac{t^2 + 7t + 12}{t + 3} =$
- A) $t + 3$
 - B) $t + 4$
 - C) $t + 7$
 - D) $t^2 + 4$
 - E) $t^2 + 7$
- 4 One solution to $x^2 + 2x - 15 = 0$ is
- A) -3
 - B) -1
 - C) 1
 - D) 2
 - E) 3
- 5 If $x = -2$ and $y = 3$, then $\frac{x^2 + xy}{x^2 - y} =$

A) -2

B) -1

C) 0

D) 1

E) 2

6) If $\frac{5}{x-1} = \frac{2}{x+2}$, then $x =$

A) -6

B) -4

C) -2

D) 2

E) 4

7) $\frac{a}{b} + \frac{b-a}{b} =$

A) 0

B) $\frac{a}{b}$

C) 1

D) $\frac{b-a}{b}$

E) $\frac{2a-b}{b}$

8) The line through $(1, 2)$ and $(5, -2)$ has equation

- A) $y = -x + 3$
- B) $y = x + 1$
- C) $y = -x - 1$
- D) $y = \frac{1}{2}x + \frac{3}{2}$
- E) $y = -2x + 4$

9) The x -intercept of $y = 4x - 20$ is

- A) -5
- B) 0
- C) 4
- D) 5
- E) 20

10) For the solution to

$$\begin{cases} 2x + 3y = 13 \\ x - y = 1 \end{cases}$$

the value of y is

- A) 1
- B) $\frac{3}{2}$
- C) 2
- D) $\frac{9}{5}$
- E) $\frac{11}{5}$

11) The line passing through $(-1, 4)$ and parallel to $y = -3x + 2$ is

- A) $y = -3x + 4$
- B) $y = 3x + 1$
- C) $y = -3x + 1$
- D) $y = -x + 3$
- E) $y = 3x - 1$

12) $(3x - y)(3x + y) =$

- A) $9x^2 + y^2$
- B) $9x^2 - y^2$
- C) $6x^2 - y^2$
- D) $9x^2 - 6xy - y^2$
- E) $9x^2 + 6xy - y^2$

13) If $3x + 7 = 16$, then $2x - 1 =$

- A) 3
- B) 4
- C) 5
- D) 6
- E) 7

14) $12ab^2 - 18a^2b + 6ab =$

- A) $6ab(2b - 3a + 1)$
- B) $6ab(2b - 3a - 1)$
- C) $6a(2b - 3a + 1)$
- D) $6b(2b - 3a + 1)$
- E) $12ab(b - a) + 6ab$

15) $(2x^2 - x + 5) - (x^2 + 4x - 3) =$

- A) $x^2 - 5x + 8$
- B) $x^2 + 3x + 2$
- C) $x^2 - 3x + 8$
- D) $-x^2 - 5x + 8$
- E) $x^2 - 5x - 8$

Part II — Exponents, Rational Expressions, and Functions

16) $(\sqrt{a} - 3)^2 =$

- A) $a - 3\sqrt{a} + 9$
- B) $a - 6\sqrt{a} + 9$
- C) $a + 6\sqrt{a} + 9$
- D) $a - 6\sqrt{a} - 9$
- E) $a + 9$

17) $\frac{7}{3a} - \frac{a}{2} =$

- A) $\frac{14 - 3a^2}{6a}$
- B) $\frac{7 - a}{6a}$
- C) $\frac{14 - a^2}{6a}$
- D) $\frac{7 - 2a}{6a}$
- E) $\frac{14 - 3a}{6a}$

18) $x^{-4} =$

- A) $-x^4$
- B) $\frac{1}{x^2}$
- C) $\frac{1}{x^4}$
- D) x^4
- E) x^{-2}

19) $\frac{5}{\sqrt{2}} =$

- A) $\frac{\sqrt{2}}{5}$
- B) $\frac{5\sqrt{2}}{2}$
- C) $\sqrt{10}$
- D) $\frac{10\sqrt{2}}{5}$
- E) $\frac{2\sqrt{2}}{5}$

20) Solve $4x + 3 \geq 3(x - 1)$.

A) $x \geq -6$

B) $x \leq -6$

C) $x \geq 6$

D) $x \leq 6$

E) $x > -6$

21) $\left(\frac{x}{x + xy}\right)\left(\frac{y^2 + y}{y^2}\right) =$

A) 1

B) $\frac{1}{y}$

C) $\frac{1}{x}$

D) $\frac{1}{x + y}$

E) $\frac{x}{y}$

22) $(-3xy^2)(x^2y)^2 =$

A) $-3x^3y^4$

B) $-3x^5y^4$

C) $3x^5y^4$

D) $-3x^5y^2$

E) $-6x^5y^4$

23) $x^{3/4} \cdot x^{5/4} =$

A) x

B) x^2

C) $x^{8/16}$

D) $x^{15/16}$

E) x^3

24) $\frac{x^{n+5}}{x^3} =$

A) x^{n+2}

B) x^{n+8}

C) x^{n-2}

D) x^{2n+2}

E) x^{n+1}

25) Given $f(x) = x^2 + 2x - 1$, find $f(a - 1)$.

- A) $a^2 - 2$
- B) $a^2 + 2$
- C) $a^2 - 1$
- D) $a^2 - 2a$
- E) $a^2 + 2a - 2$

26) $\frac{a+b}{\frac{1}{a} + \frac{1}{b}} =$

- A) $a + b$
- B) ab
- C) $\frac{ab}{a+b}$
- D) $a^2 + b^2$
- E) $\frac{a+b}{ab}$

27) If $y = \frac{3x-4}{5}$, then $x =$

- A) $\frac{5y-4}{3}$
- B) $\frac{5y+4}{3}$
- C) $\frac{3y+4}{5}$
- D) $5y+4$
- E) $\frac{5y-3}{4}$

28) $\frac{x^2 - 4x}{x - 4} =$

- A) $x - 4$
- B) x
- C) $x + 4$
- D) x^2
- E) 1

29) Solve $(x - 1)(x + 6) > 0$.

- A) $-6 < x < 1$
- B) $x < -6$
- C) $x > 1$
- D) $x < -6$ or $x > 1$
- E) $x > -6$

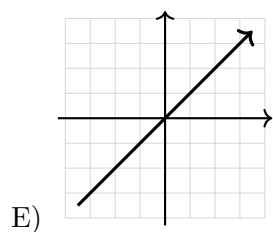
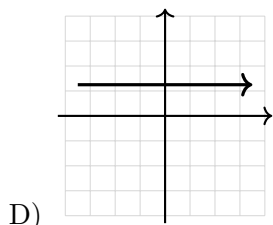
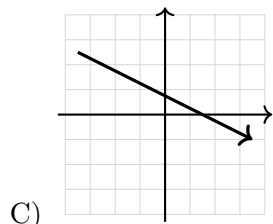
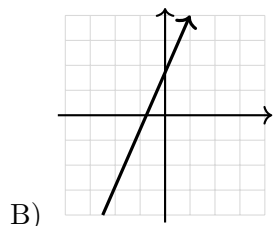
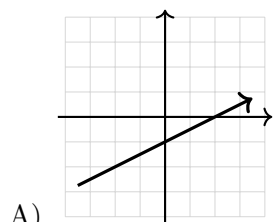
- 30 In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

$$\frac{1}{cx} = \frac{x}{76} + \frac{1}{c}$$

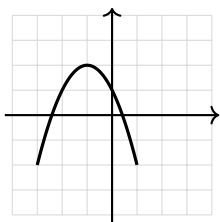
- A) 19.5
- B) -20
- C) 20
- D) -19
- E) 19

Part III — Graphs, Geometry, and Radical Functions

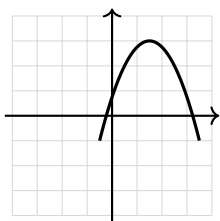
- 31 Which of the following could be the graph of $y = \frac{1}{2}x - 1$?



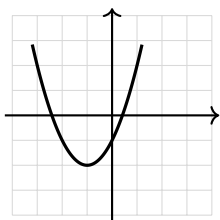
- 32 Which of the following could be the graph of $y = -(x + 1)^2 + 2$?



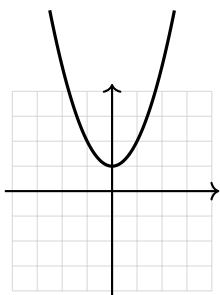
A)



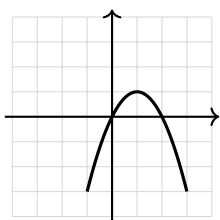
B)



C)

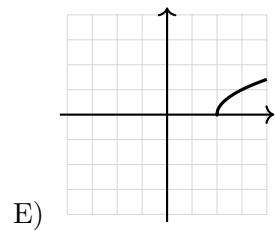
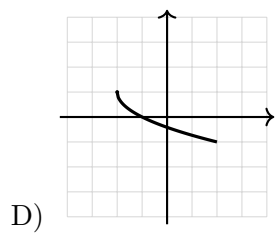
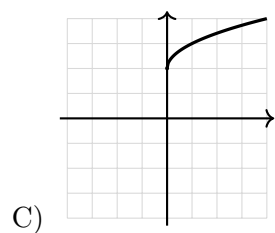
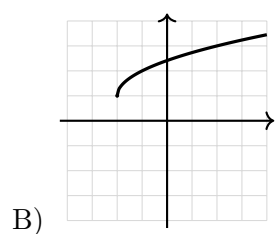
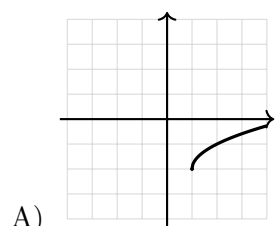


D)

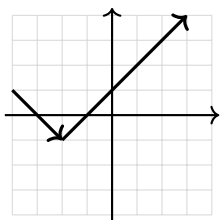


E)

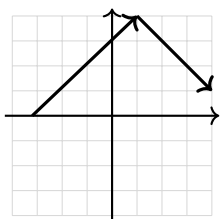
- 33 Which of the following could be the graph of $y = \sqrt{x - 1} - 2$?



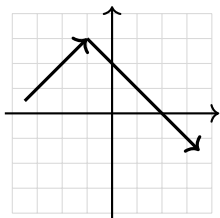
- 34 Which of the following could be the graph of $y = |x + 2| - 1$?



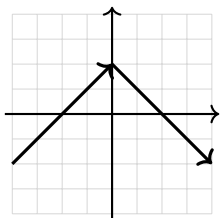
A)



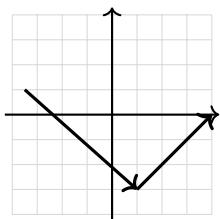
B)



C)

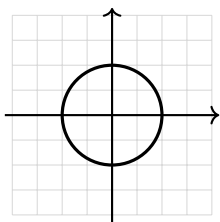


D)

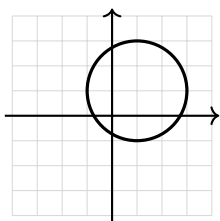


E)

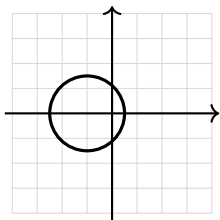
- 35) Which graph could represent the circle $x^2 + y^2 = 4$?



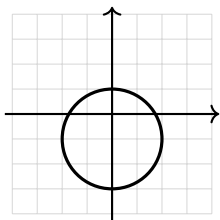
A)



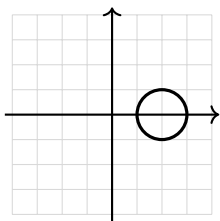
B)



C)



D)



E)

- 36) Solve $7 - \sqrt{x+5} = 4$.

- A) 2
B) 3
C) 4
D) 7
E) 9

- 37) $\sqrt{16m^2 + 16n^2} =$
- A) $4(m + n)$
 - B) $16\sqrt{m^2 + n^2}$
 - C) $4\sqrt{m^2 + n^2}$
 - D) $\sqrt{m^2 + n^2}$
 - E) $4mn$
- 38) In a triangle, two angles are 50° and 65° . The third angle is
- A) 55°
 - B) 60°
 - C) 65°
 - D) 70°
 - E) 75°
- 39) An isosceles triangle has side lengths 7, 7, and 10. Its perimeter is
- A) 17
 - B) 20
 - C) 21
 - D) 24
 - E) 27
- 40) $\log_2 32 =$
- A) 3
 - B) 4
 - C) 5
 - D) 6
 - E) 8
- 41) If $x < 2$, then $\frac{|x - 2|}{x - 2} =$
- A) 1
 - B) -1
 - C) 0
 - D) $x - 2$
 - E) $2 - x$
- 42) Solve $|x + 1| \leq 4$.
- A) $-4 \leq x \leq 4$

- B) $-5 \leq x \leq 3$
- C) $-3 \leq x \leq 5$
- D) $x \leq -5$ or $x \geq 3$
- E) $-1 \leq x \leq 4$

43 If $x \neq 2$ and

$$(x - 2)^2(x + 1) + 3(x - 2)(x + 1) = (x - 2)P,$$

then $P =$

- A) $x^2 + 2x + 1$
- B) $x^2 + x - 2$
- C) $x^2 + 3x + 2$
- D) $x^2 + 4x + 1$
- E) $x^2 - 1$

44 $\left(\frac{2x^{-1}y^2}{z}\right)^{-2} =$

- A) $\frac{4y^4}{x^2z^2}$
- B) $\frac{x^2z^2}{4y^4}$
- C) $\frac{z^2}{4x^2y^4}$
- D) $\frac{xz}{2y^2}$
- E) $\frac{4x^2y^4}{z^2}$

45 $x^{-3}(x^3 + x^7) =$

- A) x^{10}
- B) $1 + x^4$
- C) $x + x^4$
- D) $1 + x^{10}$
- E) $x^3 + x^4$

Part IV — Functions, Trigonometry, and Precalculus

46) Solve $27^{x+1} = 9^{x+2}$.

- A) -2
- B) -1
- C) 0
- D) 1
- E) 2

47) If $\log w = \log x + \log y - \log z$, then $w =$

- A) $x + y - z$
- B) $\frac{xy}{z}$
- C) xyz
- D) $\frac{xz}{y}$
- E) $\frac{z}{xy}$

48) $\sin\left(x + \frac{\pi}{2}\right) =$

- A) $\sin x$
- B) $-\sin x$
- C) $\cos x$
- D) $-\cos x$
- E) $\tan x$

49) If $\tan \theta = \frac{3}{4}$ and θ is in Quadrant I, then $\sin \theta =$

- A) $\frac{3}{4}$
- B) $\frac{3}{5}$
- C) $\frac{4}{5}$
- D) $\frac{5}{4}$
- E) $\frac{4}{3}$

50 Which function has amplitude 3 and period π ?

- A) $y = 3 \sin x$
- B) $y = \sin 3x$
- C) $y = 3 \sin 2x$
- D) $y = 2 \sin 3x$
- E) $y = 3 \cos \frac{x}{2}$

51 In triangle $\triangle ABC$, the measure of $\angle B$ is 90° , $\overline{AB}$ is a leg with a length of 12, and point D lies on segment $\overline{AC}$, and finally connect BD . Which of the following must be true?

- I. $\overline{BC} = 12 \tan(\angle CAB)$
 - II. $\cos(\angle ABD) - \sin(\angle DBC) = 0$
 - III. $\angle ABD = \angle BDC$
- A) I and II only
 - B) II and III only
 - C) I and III only
 - D) I, II, and III
 - E) None

52 The point on the unit circle corresponding to 210° is

- A) $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
- B) $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- C) $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- D) $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
- E) $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

53) The population model $P(t) = 800(1.04)^t$ means that the population

- A) Starts at 4
- B) t increases by 4 each year
- C) Grows by 4% each year
- D) Doubles every 4 years
- E) Starts at 1.04

54) Find the inverse of $f(x) = 3x - 5$.

- A) $f^{-1}(x) = \frac{x-5}{3}$
- B) $f^{-1}(x) = \frac{x+5}{3}$
- C) $f^{-1}(x) = 3x + 5$
- D) $f^{-1}(x) = \frac{3x+5}{3}$
- E) $f^{-1}(x) = \frac{x}{3} - 5$

55) Find the domain of $f(x) = \frac{\sqrt{x+1}}{x-3}$.

- A) $[-1, \infty)$
- B) $(-1, \infty) \setminus \{3\}$
- C) $[-1, 3) \cup (3, \infty)$
- D) $(-\infty, 3) \cup (3, \infty)$
- E) $[-1, 3]$

56) If $f(x) = 2x^2 - 1$, then

$$\frac{f(3) - f(-1)}{2} =$$

- A) 6
- B) 7
- C) 8
- D) 9
- E) 10

57) Which of the following is equal to $\cos 2x$?

- A) $2\cos^2 x - 1$
- B) $1 - 2\sin^2 x$
- C) $\cos^2 x - \sin^2 x$
- D) none of the above
- E) all of A, B, and C

58 In a right triangle, the hypotenuse is 13 and the side adjacent to θ is 5. Then $\sec \theta =$

A) $\frac{5}{13}$

B) $\frac{12}{13}$

C) $\frac{13}{5}$

D) $\frac{13}{12}$

E) $\frac{5}{12}$

59 Solve $2 \sin x = 1$ on $[0, 2\pi)$.

A) $x = \frac{\pi}{6}, \frac{11\pi}{6}$

B) $x = \frac{\pi}{6}, \frac{5\pi}{6}$

C) $x = \frac{\pi}{3}, \frac{2\pi}{3}$

D) $x = \frac{\pi}{6}, \frac{7\pi}{6}$

E) $x = \frac{5\pi}{6}, \frac{11\pi}{6}$

60 If $\log_4 x = -\frac{3}{2}$, then $x =$

A) $-\frac{3}{2}$

B) $\frac{1}{4}$

C) $\frac{1}{8}$

D) 8

E) $4\sqrt{4}$

Part V — Geometry, Solid Geometry, and Advanced Readiness

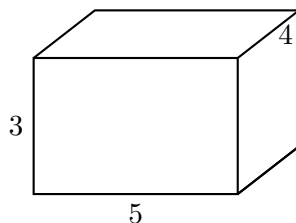
61 A right triangular pyramid has exactly six edges. Each of these edges is 96 centimeters long. If the surface area of the pyramid is $k\sqrt{3}$ square centimeters, what is the value of k ?

- A) 9216
- B) 9217
- C) 9218
- D) 9219
- E) 9220

62 Consider a square right pyramid with a volume of 400 cubic inches and a slant height of 13 inches. The pyramid's length, width, and height (in inches) are all integers. What is the surface area of the pyramid in cubic inches?

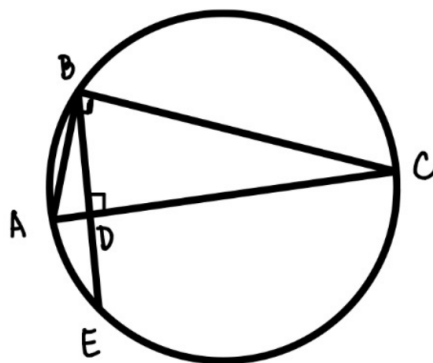
- A) 320
- B) 360
- C) 400
- D) 440
- E) 480

63 Find the volume of the rectangular prism shown.



- A) 12
- B) 20
- C) 45
- D) 60
- E) 75

- 64 In the figure shown, points A, B, C , and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D , and $BD = \sqrt{346}$. The diameter of the circle is 175. If $\frac{CD}{AD} = r$, what is the value of r ?



- A) 86.5
 B) 85.5
 C) 90
 D) 216
 E) 324
- 65 In the xy -plane, the graph of $x^2 + 6x + y^2 + 2y = 15$ is a circle. Line k has a slope of $\frac{3}{4}$ and is tangent to this circle at the point (a, b) , where b is negative. What is the value of b ?
- A) -5
 B) -4
 C) -3
 D) -2
 E) -1
- 66 The volume of a sphere with radius 4 is
- A) $\frac{64\pi}{3}$
 B) $\frac{128\pi}{3}$
 C) $\frac{256\pi}{3}$
 D) 64π
 E) 128π

- 67) Two similar triangles have side-length ratio 1 : 3. If the area of the smaller triangle is 8, the area of the larger triangle is
- A) 24
 - B) 36
 - C) 48
 - D) 72
 - E) 96
- 68) Rectangular prism A and rectangular prism B are similar. The height of rectangular prism A is 4 inches and the height of rectangular prism B is 10 inches. If the volume of prism A is 200 cubic inches, then what is the volume of prism B?
- A) 1250
 - B) 2500
 - C) 3125
 - D) 5000
 - E) 6125
- 69) The distance between $(1, -2)$ and $(5, 1)$ is
- A) 4
 - B) 5
 - C) $\sqrt{7}$
 - D) $\sqrt{17}$
 - E) 7
- 70) The equation of the circle with center $(2, -1)$ and radius 4 is
- A) $(x + 2)^2 + (y - 1)^2 = 16$
 - B) $(x - 2)^2 + (y + 1)^2 = 16$
 - C) $(x - 2)^2 + (y - 1)^2 = 16$
 - D) $(x + 2)^2 + (y + 1)^2 = 16$
 - E) $x^2 + y^2 = 16$

Directions

- Fill in exactly one answer for each question.
- Erase clearly if you change an answer.
- Total questions: **70**.
- No calculators unless specifically permitted by the instructor.

Student Answer Grid

1	25	49
2	26	50
3	27	51
4	28	52
5	29	53
6	30	54
7	31	55
8	32	56
9	33	57
10	34	58
11	35	59
12	36	60
13	37	61
14	38	62
15	39	63
16	40	64
17	41	65
18	42	66
19	43	67
20	44	68
21	45	69
22	46	70
23	47	
24	48	

Teacher Scoring Section

Part Scores

Part I (Q1–15): _____ / 15

Part II (Q16–30): _____ / 15

Part III (Q31–45): _____ / 15

Part IV (Q46–60): _____ / 15

Part V (Q61–70): _____ / 10

Overall Placement

Total Score: _____ / 70

Recommended Course:

Reviewer Notes:

Placement Interpretation Guide

Recommended Placement by Total Score

Score Range	Suggested Course
0–18	Algebra I
19–31	Geometry
32–46	Algebra II
47–60	Precalculus
61–70	Calculus Readiness

Important: Final placement should also consider pacing, work quality, topic distribution, and mathematical maturity.

Teacher Interpretation Page

How to Use This Placement Test

This assessment is designed to estimate a student's best starting point among Algebra I, Geometry, Algebra II, Precalculus, and Calculus-readiness coursework. The total score gives a strong first recommendation, but the final decision should also be based on the student's pattern of strengths and weaknesses.

Recommended process:

1. Check the total score range.
2. Review which section(s) caused the score drop.
3. Consider whether mistakes were conceptual or careless.
4. Use teacher judgment before finalizing placement.

Section-by-Section Meaning

Part I — Foundations and Algebra Skills (Q1–15) Measures factoring, equations, lines, systems, algebraic simplification, and core symbolic fluency.

Part II — Exponents, Rational Expressions, and Functions (Q16–30) Measures exponent rules, rational expressions, function substitution, inverse-style algebra, and abstract manipulation.

Part III — Graphs, Geometry, and Radical Functions (Q31–45) Measures graph recognition, transformations, radicals, absolute value, logs, and basic geometric reasoning.

Part IV — Functions, Trigonometry, and Precalculus (Q46–60) Measures trigonometric identities, function models, inverse functions, domain analysis, and precalculus readiness.

Part V — Geometry, Solid Geometry, and Advanced Readiness (Q61–70) Measures area, volume, circle geometry, similarity, spatial reasoning, and late-stage readiness.

Detailed Placement Notes**0–18: Algebra I**

Student likely needs support with core algebra skills such as solving equations, factoring, linear relationships, and symbolic manipulation. This range usually indicates that moving directly into Geometry or Algebra II would be too fast.

19–31: Geometry

Student shows partial algebra readiness but may still have inconsistency in symbolic work. Geometry is often appropriate when basic algebra exists but abstract function work and higher-level manipulation are not yet stable.

32–46: Algebra II

Student demonstrates reasonable fluency with foundational algebra and some success with functions, exponents, and rational expressions. However, graph transformations, trigonometry, and higher-level function analysis may still be too weak for Precalculus.

47–60: Precalculus

Student is generally strong in algebra and functions, and can handle a meaningful portion of the trigonometry and advanced topics. Precalculus is the most appropriate course when the student is not yet consistently strong enough across all higher-level topics to be considered fully Calculus-ready.

61–70: Calculus Readiness

Student demonstrates strong command of algebra, functions, graph analysis, trigonometry, and geometry. This range suggests readiness for a calculus-track course, assuming school grades, work quality, and pacing also support that recommendation.

Teacher Comment Box

Strengths:

Areas to Review:

Final Placement Recommendation:

Answer Key

- | | | | | |
|--------------|--------------|--------------|--------------|--------------|
| 1. <i>A</i> | 2. <i>E</i> | 3. <i>B</i> | 4. <i>E</i> | 5. <i>A</i> |
| 6. <i>B</i> | 7. <i>C</i> | 8. <i>A</i> | 9. <i>D</i> | 10. <i>E</i> |
| 11. <i>C</i> | 12. <i>B</i> | 13. <i>C</i> | 14. <i>A</i> | 15. <i>A</i> |
| 16. <i>B</i> | 17. <i>A</i> | 18. <i>C</i> | 19. <i>B</i> | 20. <i>A</i> |
| 21. <i>B</i> | 22. <i>B</i> | 23. <i>B</i> | 24. <i>A</i> | 25. <i>A</i> |
| 26. <i>B</i> | 27. <i>B</i> | 28. <i>B</i> | 29. <i>D</i> | 30. <i>D</i> |
| 31. <i>A</i> | 32. <i>A</i> | 33. <i>A</i> | 34. <i>A</i> | 35. <i>A</i> |
| 36. <i>C</i> | 37. <i>C</i> | 38. <i>A</i> | 39. <i>D</i> | 40. <i>C</i> |
| 41. <i>B</i> | 42. <i>B</i> | 43. <i>A</i> | 44. <i>B</i> | 45. <i>B</i> |
| 46. <i>D</i> | 47. <i>B</i> | 48. <i>C</i> | 49. <i>B</i> | 50. <i>C</i> |
| 51. <i>A</i> | 52. <i>B</i> | 53. <i>C</i> | 54. <i>B</i> | 55. <i>C</i> |
| 56. <i>C</i> | 57. <i>E</i> | 58. <i>C</i> | 59. <i>B</i> | 60. <i>C</i> |
| 61. <i>A</i> | 62. <i>B</i> | 63. <i>D</i> | 64. <i>A</i> | 65. <i>A</i> |
| 66. <i>C</i> | 67. <i>D</i> | 68. <i>B</i> | 69. <i>B</i> | 70. <i>B</i> |